

9 Seminar: Strategies and management in Race Track engineering

9.1 Introduction

This seminar delineates the technical, managerial, and psychological frameworks required for success in elite motorsport, based on insights from Eng. Marco Fainello's professional experience in Formula 1 and Endurance racing. The content shifts focus from traditional academic equations toward the practical integration of team dynamics, technological tools like Artificial Intelligence, and the operational distinction between testing and performance environments. It is posited that professional mastery in this field requires a synthesis of technical competence and soft skills often omitted from standard university curricula.

9.2 The Competitive Environment and Team Synergy

Motorsport is fundamentally defined by relentless competition, where losing is statistically a more frequent occurrence than winning. Professional growth depends on the ability to process the frustration of defeat and utilize it as a mechanism for improvement. A critical distinction is made between positive and negative competition. Positive competition acts as an internal drive that forces colleagues to improve together. Conversely, negative competition is a destructive dynamic where individuals withhold data or undermine others to highlight personal success, a phenomenon often observed in high-pressure environments like Formula 1. True professional excellence is achieved through "clever selfishness," recognizing that individual rewards are only attainable through the collective success of the team.

9.3 Technological Integration: Engineering vs. Data Science

The adoption of Artificial Intelligence and Machine Learning is viewed as a supplementary tool rather than a replacement for traditional engineering. The most effective approach involves integrating neural networks and optimization tools with a deep understanding of mechanical and electronic systems. Furthermore, data must be balanced with the "human sensor"—the driver. While telemetry provides objective precision, a driver often senses vehicle instabilities or track surface changes before they are reflected in the data streams.

9.4 Strategic Differences: Formula 1 and Endurance Racing

Two distinct operational philosophies are identified within the industry. Formula 1 is characterized as a "perfect organization" focused on forecasting and performance, with highly standardized communication protocols. Endurance racing is constrained by results and efficiency rather than raw power. Success in endurance is predicated on resilience, the management of unexpected variables over long durations (e.g., 24 hours), and deeper interpersonal relationships between engineers and multiple drivers sharing a single car.

9.5 Managerial Leadership and the Orchestra Model

The role of a manager is compared to that of an orchestrator, responsible for decomposing a macro-goal (e.g., winning a race) into measurable micro-goals for specific departments.

A clear separation is maintained between leadership and authority. Leadership involves coordinating activities and ensuring team members are motivated within their roles. Authority is derived from technical wisdom and experience; the individual with the most authority in a room is often a senior technician rather than the formal manager. A recurring bottleneck in management is the failure to delegate, as engineers often believe they can complete tasks faster themselves. However, delegation is a vital investment that frees mental energy for higher-level problem-solving.

9.6 Operational Methodologies: Test vs. Performance Modes

Professional activities are categorized into two distinct modes. Test Mode is a learning-focused environment where making "stupid mistakes" is encouraged to define the limits of the system. Performance Mode is the execution phase where risks must be minimized. In this mode, engineers and drivers must operate with a "margin" to ensure a result is secured, potentially sacrificing peak performance for reliability.

9.7 Cultural Awareness and Communication

In multicultural environments, communication is often the primary limitation to performance. What is considered "obvious" varies significantly across cultures, leading to misunderstandings even when a technical language is shared. Effective communication requires proactive verification and the courage to ask for clarification to minimize damage caused by cultural and linguistic gaps.

9.8 Strategic Preparation and Failure Analysis

Success is defined as the intersection of preparation and opportunity. High-level engineering requires "thinking before the problem happens" by conducting exhaustive risk analyses. This involves creating fallback strategies for every possible sensor or actuator failure. Reacting only to recent problems is identified as a losing strategy; instead, resources must be prioritized toward what is critical for the result rather than what is merely interesting.

9.9 Professional Philosophy: Work-Life Integration

The concept of "work-life balance" is challenged in favor of work-life integration. It is argued that professional choices are inseparable from personal identity. Success in this demanding field requires total commitment, the courage to adapt to changing regulations, and the proactive pursuit of collaboration.

10 Seminar: Racing Tires

10.1 Intro - Main Differences Between Racing and Road Tires

This section outlines the primary differences between racing and road tires. Analyzing the physical quantities and stress levels reveals massive differences, highlighting completely distinct usage profiles. To better understand these profiles, we can look at the core design philosophy for both applications.

10.1.1 Design Philosophy and Construction

- **Racing Tires:** These are designed specifically to optimize handling, grip, and overall lap time.
 - They feature a slick tread with no grooves or cuts, which maximizes the contact area with the track.
 - The rubber compound is engineered to maximize grip within a specific, restricted temperature range of 80-120°C.
 - To ensure steering precision and a fast response, the sidewall is designed with high stiffness.
- **Passenger Car Tires:** These require a compound trade-off to balance wet grip, rolling resistance, and wear.
 - They utilize tread patterns and grooves to divide stiffness and evacuate water.
 - Unlike racing tires, passenger sidewalls are primarily optimized to protect against sidewalk impacts.



Figure 150 Racing



Figure 151 Passenger Car Tire

10.1.2 Competitive racing environment

In competitive racing, a single tire manufacturer often supplies the entire grid. Because the hardware is identical for everyone, the competitive advantage lies in understanding how to use the tire better or earlier than the competition. This leads to a constant evolution in how tires are managed over a race weekend:

- Teams develop customized usage strategies based on their specific vehicle, the driver, and event circumstances.
- Tire providers collect telemetry data after events to find performance correlations, compare simulations against reality, and improve the accuracy of their virtual models.

To assist the teams, manufacturers provide track support engineers to ensure they operate the tires within strict safety and performance prescriptions. Specifically, two critical parameters are heavily monitored to prevent failures and maximize performance:

- **Maximum Camber Angle:** While a higher negative camber yields greater lateral force, it must be limited to manage temperature and durability, preventing the inner rim side from overloading and blistering.
- **Minimum Inflation Pressure:** Lower pressure provides more grip, but a minimum amount of air is strictly required to maintain the tire's load-carrying capacity.

10.2 Basic Tire Shape Design

Moving from the general requirements to the actual geometry, the outer profile of the tire dictates the shape of the contact patch, which in turn influences the vehicle's dynamic behavior.

Profile Shape and Footprint

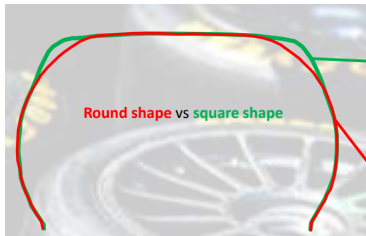


Figure 152 Profile



Figure 153 squared footprint

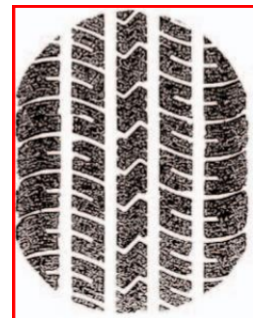


Figure 154 rounded footprint

- **Square Profile:** Produces a squarer footprint, yielding better handling, dry traction, and uniform wear. The external ribs engage directly, reaching peak lateral force (F_y) at lower slip angles.
- **Round Profile:** Creates a rounder footprint, favored for wet conditions and reducing pattern noise. External ribs engage gradually, delaying the F_y peak to larger slip angles.

Equilibrium Shape and Tension Control

An inflated tire naturally seeks an "equilibrium shape" characterized by uniform carcass tension and no bending. However, because the tire distorts under load, designers impose an initial shape with non-uniform tension that better mimics the deflected state.

The relationship between internal pressure, shape, and tension is governed by the Laplace equation. By altering the meridian radius of curvature (r_1), designers can control local stiffness across different areas of the tire:

- **Near the Belt:** A smaller r_1 lowers carcass tension, providing more shoulder flexibility. Concurrently, a smaller angle ($\sin \theta$) increases belt tension to prevent buckling and improve contact pressure during cornering.
- **Near the Bead:** A larger r_1 increases carcass tension, creating a stiffer bead area that enhances both handling and durability.

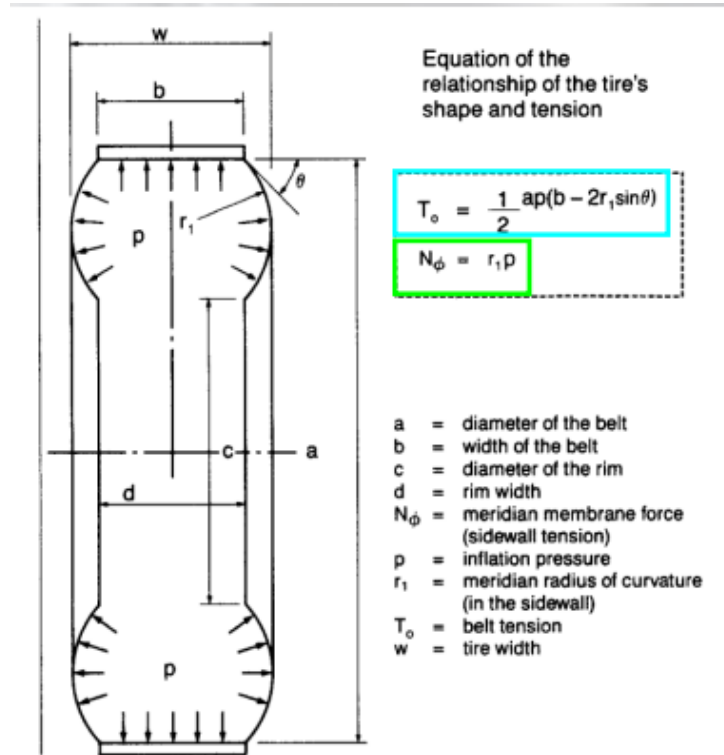


Figure 155

Crown Optimization

By carefully balancing these variables, the crown shape is optimized to minimize pressure deviation across the contact patch. This ensures a uniform load distribution, which is especially critical in the shoulder area.

10.3 Rubber Compound: Viscoelastic Properties

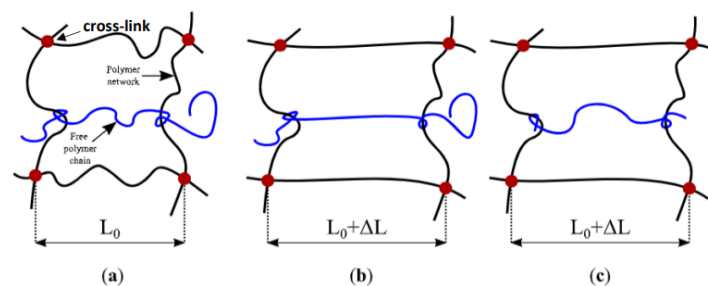


Figure 156

Composition and Structure

Rubber compounds blend elastomers, which form the structural backbone, with fillers like Carbon Black and Silica to increase stiffness and tear resistance. Sulfur cross-links polymer chains during vulcanization, while specific resins and oils are added to extend grip performance across high-temperature ranges. This unique chemical structure leads to complex physical behaviors under dynamic loads.

Viscoelastic Behavior

Because it is not a perfectly elastic material, rubber responds to loads with a mix of elastic and viscous properties:

- **Moduli:** Rubber exhibits both elastic behavior (Storage Modulus, E') and viscous behavior (Loss Modulus, E''). In racing compounds, carbon black dissipates more energy at high temperatures, which translates to a smaller drop in the G' modulus as temperature increases.

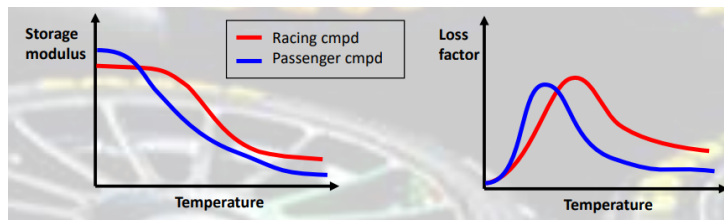


Figure 157

- **Hysteresis ($\tan \delta$):** The ratio E''/E' measures the material's damping, or energy loss. Racing compounds rely on specific resins to widen this $\tan \delta$ peak at high temperatures, ensuring sustained grip under extreme stress.
- **Dynamic Softening:** Rubber stiffness dynamically decreases with increasing strain amplitude (the Payne effect) and with repeated loading cycles (the Mullins effect).

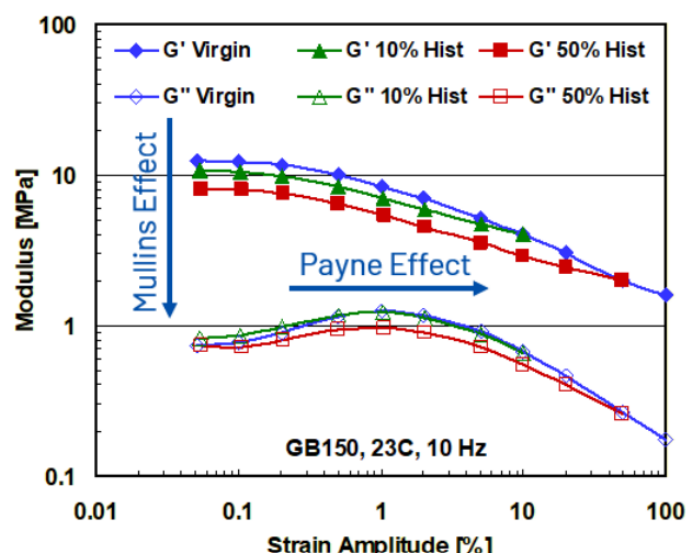


Figure 158

10.4 Tire Track Data Analysis

Grip and Pressure Evolution

On the track, understanding how the tire behaves dynamically is just as important as its design. Telemetry is used to track several performance indicators over time:

- **Grip Factors:** Telemetry isolates braking, traction, cornering, and aerodynamic grip to evaluate the overall combined grip: $G_{comb} = \sqrt{G_{Lat}^2 + G_{Long}^2}$.
- **Warm-up vs. Long Runs:** On-board TPMS sensors track both inflation pressure and cavity air temperature. Because pressures typically take about 8 laps to stabilize, evaluating vehicle balance strictly during the warm-up phase can lead to incorrect setup decisions for longer stints.

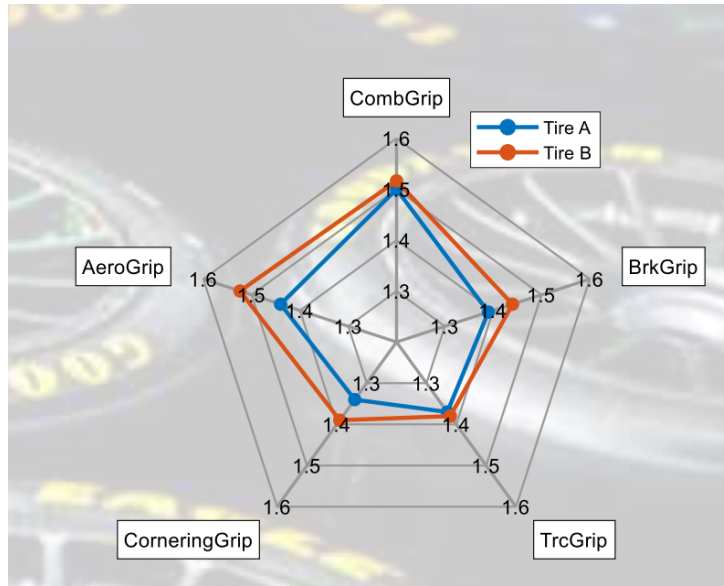


Figure 159

To properly evaluate this balance during the transient phases, engineers look at the Inflation Pressure (IP) Ratio and Temperature Ratio between the front and rear axles:

$$\text{IP Ratio} = \frac{FL + FR}{FL + FR + RL + RR}$$

$$\text{Temperature Ratio} = \frac{T_{FL} + T_{FR}}{T_{FL} + T_{FR} + T_{RL} + T_{RR}}$$

During the warm-up, these ratios often show a slightly higher front bias. It is only when the tire parameters plateau that the true dynamic balance of the car can be reliably assessed.

Temperature Ranges and Camber Evaluation

Beyond the internal air temperature, the surface temperature of the tread is fundamentally tied to the chemical grip generated.

- **Operating Windows:** Tires only generate peak grip within specific thermal boundaries. Track engineers monitor surface temperature efficiency to ensure the tires spend maximum time within their ideal thermal ranges (e.g., 80°C to 95°C for front

racing slicks). If the tire operates below or above this window, performance drops significantly.

- **Camber Optimization:** Multi-channel temperature sensors measure the thermal spread across the tread width (Inside, Mid, Outside) corner by corner. Analyzing the temperature differential between the inner and outer shoulders helps teams verify if the camber angle is optimized for the specific track layout.

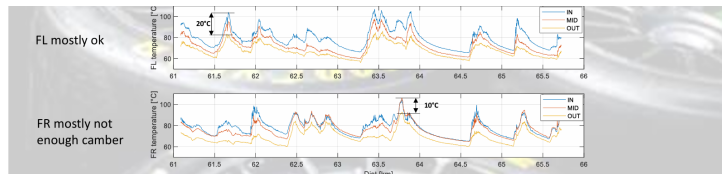


Figure 160

10.5 Tire Track Data Analysis

Degradation Phenomena: Blistering and Graining

When tires are pushed outside their ideal operating windows or subjected to incorrect setups, two primary forms of degradation can severely impact performance:

- **Blistering:** This occurs when the core (bulk) of the tread overheats. The extreme internal heat forms air pockets that expand and burst through the surface, creating holes or chunks from the shoulder to the middle of the tire. The solution usually involves reducing the tread bulk temperature or running a stiffer, more heat-resistant compound.
- **Graining:** This happens when the tread is stressed beyond its tensile limit, often due to high lateral sliding when the tire is too cold. The surface rubber is torn away, creating small rolls of rubber (grains) that stick back onto the tread, reducing the effective contact patch and overall grip.



Figure 161 Blistering



Figure 162 Graining

11 Seminar: Sustainability in Motorsport (Marelli)

11.1 Introduction

This seminar demonstrates how sustainability serves as a primary driver of innovation in advanced motorsport engineering. By highlighting the industry's role as a "technological laboratory," it illustrates how "Design for Sustainability" (DfS) principles transform racing efficiencies into sustainable mobility solutions for the broader automotive and motorsport sectors.



Figure 163

11.2 The concept of sustainability

The modern framework of sustainability was established by the United Nations in 1987, defined as "meeting the needs of the current generation without compromising the ability of future generations to meet their own needs." This mission is built upon three essential pillars: environmental protection, social equity, and economic viability.

To provide a concrete roadmap for global development, the UN identified 17 Sustainable Development Goals (SDGs) which address a wide range of issues from poverty and inequality to climate action and industrial innovation. These goals serve as the primary reference for governments and corporations to align their strategies with global sustainability needs.



Figure 164: UN 17 Sustainable Goals

In the corporate sector, these aspirations are managed through Environmental, Social, and Governance (ESG) criteria. While sustainability represents the high-level strategy, ESG provides the measurable KPIs used for execution and accountability. Transparent reporting is vital; without it, companies risk "greenwashing," or using marketing to project an eco-friendly image that is not supported by substantive actions.

11.3 Environmental Metrics and Corporate Frameworks

The environmental impact of human activity is measured by the balance between the ecological footprint—the area required to produce resources and absorb waste—and bio-capacity—the planet's ability to regenerate those resources. Currently, humanity is in an "ecological deficit," consuming resources faster than the Earth can replenish them. This is evidenced by "Earth Overshoot Day," which in 2025 fell on July 24th. In Italy, the resource consumption rate is so high that if the entire world lived similarly, 2.9 planets would be required.

To combat climate change and align with the Paris Agreement targets (limiting warming to 1.5°C or 2°C), companies utilize the Greenhouse Gas (GHG) Protocol. This protocol is the global standard for carbon accounting and classifies emissions into three distinct "Scopes" based on their source.

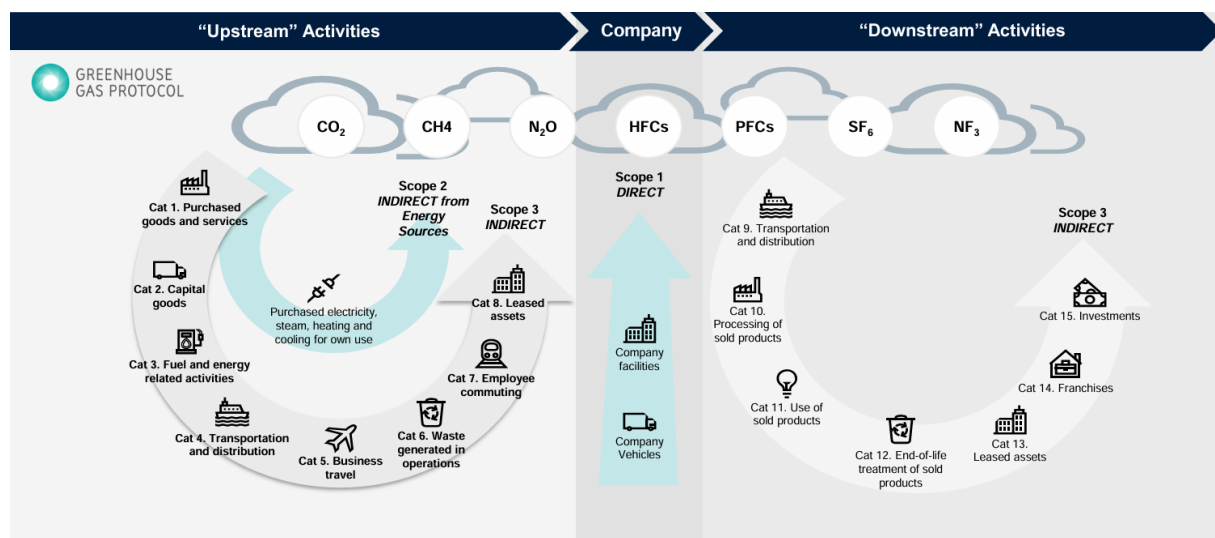


Figure 165: GHG protocol

- **Scope 1:** Direct emissions from company-owned sources (e.g., test benches, company vehicles).
- **Scope 2:** Indirect emissions from the generation of purchased electricity or heating.
- **Scope 3:** All other indirect emissions in the value chain, including purchased goods, logistics, and product end-of-life.

Understanding these scopes is critical for distinguishing between "Carbon Neutrality," which often relies on offsetting emissions through financial certificates, and "Net Zero," which requires the total elimination of emissions across the entire supply chain.

11.4 The Paradox of Motorsport: Perception vs. Reality

Motorsport is frequently criticized as an industry that "burns fuel for fun," leading to a perception of inherent unsustainability. However, data-driven analysis reveals that the direct environmental impact of the racing itself is remarkably small compared to the operations of the sport as a whole.

Using the 2022 FIA Formula 1 World Championship as a case study, data shows that race fuel emissions accounted for less than 1% of the total carbon footprint of the series. The true challenges lie in the logistics of moving the "circus" around the globe and the energy consumed by factories and facilities.

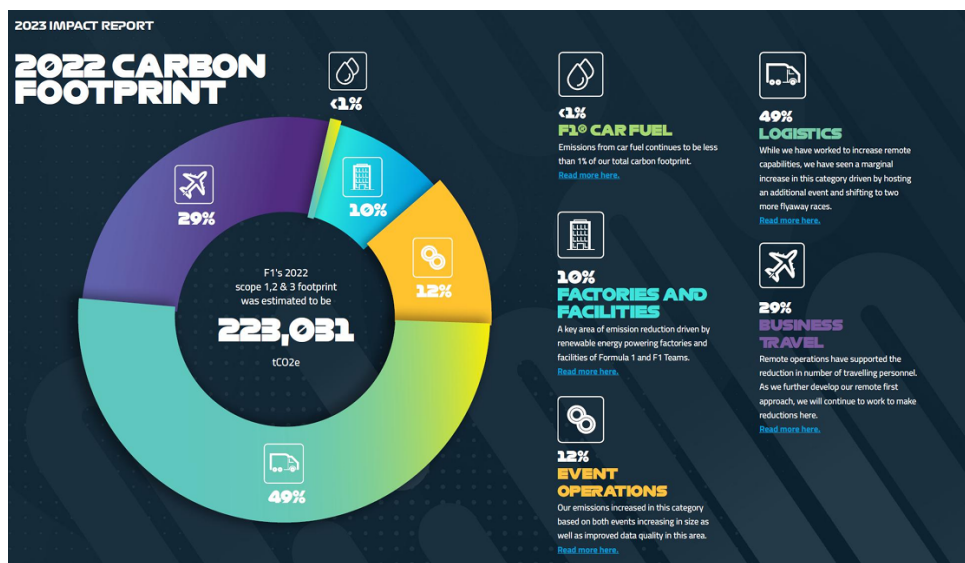


Figure 166: F1 Carbon Footprint report

This evidence confirms that for a motorsport supplier like Marelli, sustainability efforts should not focus solely on reducing race-day emissions, which are quantitatively negligible, but on using the sport as a high-speed laboratory for systemic innovation.

11.5 Innovation as a Sustainability Driver

Motorsport serves the automotive industry as a "technological laboratory." The extreme pressure of competition forces engineers to maximize efficiency—achieving the highest possible performance from a limited, regulated amount of energy. This pursuit of efficiency has historically led to numerous innovations that now define sustainable road travel:

- **Electrification:** High-performance hybrid powertrains and regenerative braking systems (KERS/ERS).
- **Materials:** The development of lightweight carbon fiber and advanced composites to reduce vehicle mass.
- **Fuels:** The current transition toward 100% sustainable synthetic fuels.
- **Control Systems:** Advanced telemetry and data analysis used to optimize engine and battery management.

11.6 Design for Sustainability (DfS) Case Studies

To integrate these concepts into engineering practice, Marelli uses a "Design for Sustainability" (DfS) methodology. This approach evaluates every stage of a product's life, from material sourcing and manufacturing to usage and end-of-life.

11.6.1 Carbon-Fiber Chassis

Traditional carbon fiber is energy-intensive to produce and nearly impossible to recycle due to thermosetting resins. Sustainable engineering solutions include the use of natural fibers (such as linen) for non-structural parts and the development of bio-based resins. Furthermore, "Design for Disassembly" allows for modular chassis components that can be repaired or refurbished rather than replaced.

11.6.2 Electronic and Electromechanical Units

For components like Vehicle Control Units (VCUs) and Motor Generator Units (MGUs), the focus is on reducing the use of rare earth metals (like neodymium) and toxic chemicals used in PCB manufacturing. Strategies include:

- **Modular Design:** Allowing for the replacement of a single broken board instead of the entire unit.
- **Circular Economy:** Implementing "second life" programs where components used in Formula 1 are refurbished for less demanding series like Formula 2.
- **Alternative Materials:** Investigating ferrite or alnico magnets to reduce reliance on high-impact mining.

11.6.3 Fuel Injectors

The engineering of fuel injectors focuses on extreme precision and durability under harsh operating conditions. Current challenges include the use of high-grade alloys and energy-intensive machining processes that generate significant material waste. Sustainable solutions include:

- **Material Optimization:** Utilizing recycled stainless steel or aluminum where performance remains unaffected and adopting bio-based polymers for non-structural components like seals and covers.
- **Remanufacturing and Reuse:** Implementing programs where retired injectors are cleaned, tested, and refitted for use in different championships, effectively extending the product's lifecycle.
- **Modular Subcomponents:** Designing the unit so that specific parts—such as the needle, solenoid, or body—can be replaced or recycled individually rather than disposing of the entire injector.

11.6.4 Racing Tires

Tires represent a significant environmental challenge due to their very short lifespan in racing and the complexity of their chemical composition. The transition toward sustainable tires involves:

- **Sustainable Sourcing:** Increasing the use of certified natural rubber and reducing reliance on petroleum-derived synthetic compounds.
- **Circular Material Use:** Incorporating recycled rubber powder from end-of-life tires into new compounds and implementing closed-loop manufacturing to recover production scrap.
- **Specialized Recycling:** Partnering with specialist facilities to repurpose worn racing tires into materials for athletic surfaces, pavement additives, or shoe soles, preventing them from being incinerated or landfilled.

11.7 Conclusion

In advanced motorsport engineering, sustainability is not a secondary marketing goal but a core engineering challenge. By leveraging the industry's unique position as an innovation hub, engineers can develop the materials, processes, and efficiency standards that will transition the global automotive sector toward a net-zero future. Success in this field requires evidence-based decision-making and a commitment to integrating ESG principles into every design choice.

12 Seminar: F1 Vehicle Setup (Ferrari)

12.1 Introduction and Overview

The seminar provided a comprehensive overview of the extreme complexity involved in optimizing a modern Formula 1 vehicle during a race weekend. The core theme revolved around managing the interplay between multiple sophisticated subsystems (such as aerodynamics, tires, Power Unit, brakes, and suspensions) which must work together in harmony to maximize track performance.



Figure 167

12.2 Car Design vs. Race Weekend

The engineering workflow in Formula 1 is divided into two distinct macro-phases with different constraints:

The **Car Design** phase aims to build the theoretically quickest car; engineers have more time, fewer constraints, and a vast array of parameters to modify while interpreting the technical regulations.

The **Race Weekend** focuses entirely on setup adjustment to extract the ultimate performance from the existing design. During the weekend, changes are highly limited.

Because all teams adopt similar deep optimization methodologies, finding even a marginal advantage requires immense effort, precise data analysis, and predictive modeling.

12.3 The Challenge of Trade offs: Ground Effect Regulation

Finding the quickest car requires a deep, iterative optimization process. A central point of the seminar was the description of the "short blanket" effect: improving performance in one area almost inevitably compromises another.

One of the main examples of this trade-off is the interaction between aerodynamics and vehicle dynamics in previous years' ground effect cars (2022 - 2025 regulations):

- **Aero/Mechanical Conflict:** These cars must run at extremely low ride heights to maximize the aerodynamic downforce coefficient.

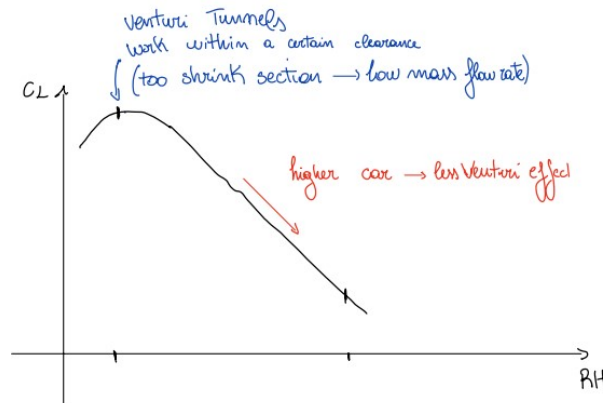


Figure 168

However, this increases the risk of the chassis hitting the ground (bottoming). If bottoming occurs, the vertical force no longer passes through the tires, causing a massive loss of grip.

Moreover, to maintain the car always in the optimal aerodynamics window and to prevent any bottoming, a stiffer setup of the suspensions must be adopted. Unfortunately, this rigidity compromises the car's mechanical grip and vertical dynamics, making it difficult to ride on kerbs.

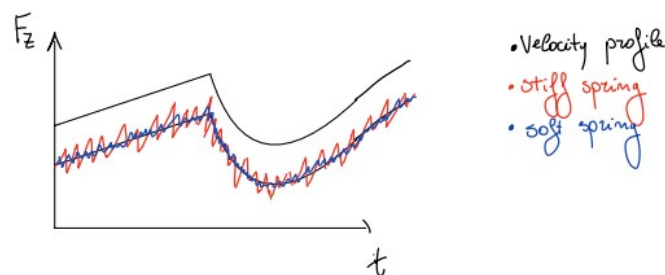


Figure 169

Mitigation of Forces fluctuations can be done with complex dampers settings.

12.4 Key Setup Tuning Parameters

To navigate the trade-offs described above and achieve specific performance targets, engineers can tune several key parameters:

- **Wings and Aero Maps:** Adjusting Front and Rear wing levels (FW and RW angles) directly alters the overall aero balance and total downforce, adapting the car to track-specific requirements
- **Suspension Stiffness:** Modifying front and rear heave (vertical) stiffness and roll stiffness (via anti-roll bars) is fundamental to control both the aerodynamic platform stability and the mechanical compliance in corners.

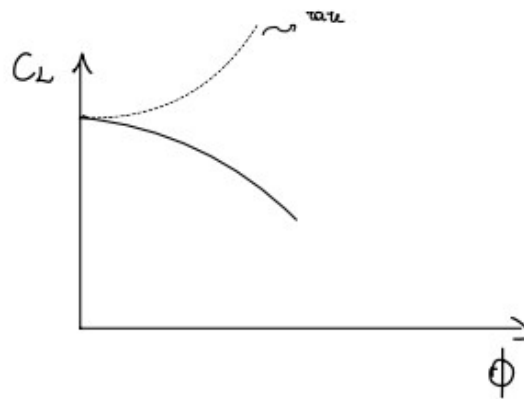


Figure 170

- **Ride Heights (FRH & RRH):** Crucial for maximizing the ground effect, but strictly limited by the risk of aerodynamic stall or chassis bottoming.

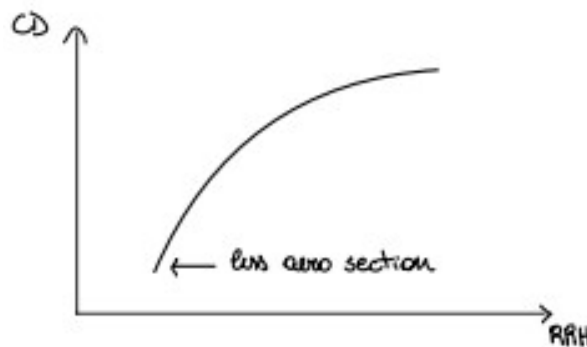


Figure 171

- **Weight Distribution (WD%):** Impacts the static vertical load on each axle, heavily affecting load transfers, traction, and overall handling balance (understeer/oversteer).
- **Tires (Toe, Camber, Pressures):** Toe and Camber are adjusted to maximize cornering grip but are constrained by strict reliability limits imposed by the tire manufacturer. Tire pressures are generally kept at the minimum allowed to maximize the contact patch.

- **Cooling:** The overall aerodynamic efficiency is also dictated by the cooling requirements of the PU, gearbox, and brakes; larger cooling openings inevitably reduce the aerodynamic surfaces available for pure performance.
- **Dampers Settings**
- **Energy Deployment Strategy**

12.5 Virtual Simulation Models

Given the limited testing time, predictive virtual models are strictly necessary to quickly analyze and correct the car setup before hitting the track. The seminar highlighted two main approaches to lapttime simulation:

- **QSS Lapttime Model (Quasi-Static Simulation):** This traditional model computes performance along pre-calculated arcs of varying curvature, always keeping the car at the theoretical edge of its performance envelope. While computationally fast, it lacks the ability to accurately simulate complex transient dynamics.
- **Dynamic Integration Model:** An advanced evolution of the QSS that performs a dynamic, progressive integration of vehicle states. It accounts for local effects (like camber variations) and transient behaviors.

12.6 Data Acquisition and Handling Stability

To refine the vehicle's performance and validate simulation models, a sophisticated data acquisition system is utilized: this includes wheel speed sensors, six-axis accelerometers, and yaw rate sensors to monitor the car's orientation and movement in real-time. Also Potentiometers and Load Cells are mounted on suspension components to measure mechanical loads.

These parameters are also extracted in order to check the stability: some data can be used to asses the vehicle under steering/over steering tendency by means of simple handling models like the single-track one. The car can be treated as a closed-loop system where the driver acts as the primary control element, stabilizing the vehicle as it operates at the edge of the performance envelope is one of the main aspects.

13 Seminar: Ducati corse

13.1 Introduction

This report details the vehicle dynamics methodologies employed by the racing department to optimize prototype motorcycle performance. Analyzing the system requires solving a highly complex dynamic problem where the rider constitutes roughly one-third of the total vehicle mass. By bridging Virtual Environment simulations (Multibody, Lap Time Simulation, aerodynamic mapping) with Real Environment telemetry (Kalman filtering for state estimation), engineers can accurately predict performance envelopes, optimize Electronic Control Unit (ECU) strategies, and define precise targets for track setups.



Figure 172Ducati MotoGP

13.2 System Complexity: The Motorcycle-Rider Equilibrium

Unlike automotive vehicle dynamics, motorcycle dynamics are heavily governed by the rider's physical mass and posture. A modern prototype motorcycle weighs approximately 160 kg, while the rider weighs around 70 kg. As the rider moves dynamically during braking, cornering, and acceleration, they fundamentally alter the Center of Gravity (COG) and the equilibrium of the entire system (roll and pitch dynamics).

Furthermore, the rider's movements dictate the aerodynamic pressure field. At high speeds, the frontal surface area and resulting aerodynamic forces change drastically depending on whether the rider is tucked behind the fairing or braking upright. Consequently, separating the motorcycle from the rider in simulations is impossible; they must be modeled as a single, highly coupled dynamic system.

13.3 Virtual Environment: Subsystem Modeling

To manage this complexity, the vehicle is decomposed into distinct virtual subsystems.

13.3.1 Rigid Body and Inertial Parameterization

The core virtual model is a multi-body system comprising the chassis, swingarm, forks, and engine. While geometric parameters are extracted directly from CAD designs, measuring the inertia tensors and precise COG of complex physical assemblies (such as the engine or carbon-fiber swingarms) is challenging. Engineers rely on specific indoor pendulum test rigs to experimentally measure the inertial properties of these subsystems, feeding the empirical data back into the multi-body model to ensure accurate baseline behavior.



Figure 173Multibody

13.3.2 Aerodynamics and Tire Dynamics

Aerodynamic modeling faces severe challenges due to extreme operational conditions. Motorcycles lean up to 60 degrees, inducing massive variations in the aerodynamic pressure fields compared to a zero-lean state. Engineers map these parameters using Computational Fluid Dynamics (CFD) and specialized wind tunnels equipped with moving belts. Because human test riders cannot sustain extreme lean postures statically for hours, detailed mannequins are utilized to extract the six aero coefficients (drag, lift, side forces, and moments) across varying pitch, yaw, and roll angles.

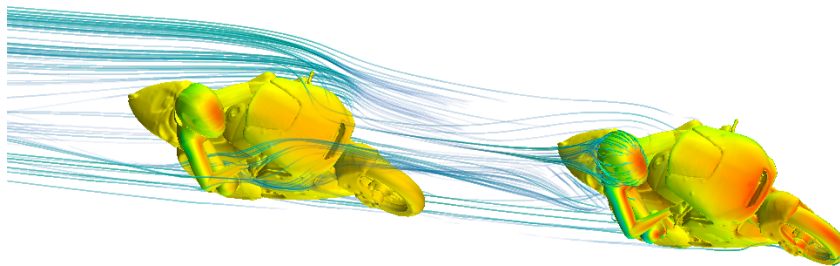


Figure 174Moto GP CFD

Tire modeling presents another hurdle. Standard tire models (such as Pacejka's Magic Formula) were developed for low-camber automotive applications. These formulas must be heavily modified and parameterized to mimic the extreme camber thrust and behavior of prototype motorcycle tires.

13.4 Simulation Methodologies

Once the subsystems are modeled, they are integrated into broader simulation environments to predict and optimize performance.

13.4.1 Full Multi-Body Simulation

Full dynamic simulations are rarely used to predict absolute lap times, as programming an artificial controller to mimic a human rider over a full lap is highly unreliable. Instead, full multi-body models are utilized for:

- **Load Estimation:** Calculating stress on specific frame components to drive Finite Element (FE) analysis.
- **Vibration Analysis:** Studying mid-to-high frequency instabilities and chassis flexibility.
- **ECU Co-Simulation:** Prototyping closed-loop strategies, such as anti-wheelie or traction control, by simulating the interaction between the ECU's PID controllers and the vehicle's torque delivery.
- **Drivetrain Dynamics:** Simulating chain tension and forces, particularly during the activation of dynamic rear ride-height devices which drastically alter the distance between sprockets.

13.4.2 Lap Time Simulation (LTS)

To determine optimal vehicle setups, Lap Time Simulation is deployed. Utilizing proprietary collocation methods, LTS ignores the closed-loop rider controller and instead searches for the absolute physical performance envelope of the motorcycle. By processing the track layout (altimetry, banking, corners), the simulation generates optimal speed and acceleration profiles. This allows engineers to evaluate the theoretical lap time benefits of varying COG positions, aerodynamic packages, or engine torque curves before the physical prototype even touches the track.

13.5 Real Environment: Telemetry and State Estimation

A model is only useful if it correlates with real-world track behavior. Bridging the gap requires advanced sensor management and mathematical state estimation.

13.5.1 The Sensor Weight Constraint

Due to strict minimum weight regulations, race motorcycles cannot carry heavy, high-fidelity instrumentation (such as optical speed sensors, lasers, or instrumented wheels) during official race weekends. The standard sensor suite is limited to an Inertial Measurement Unit (IMU), suspension stroke potentiometers, angular wheel speeds, and basic engine sensors. Therefore, algorithms must be developed to extract complex dynamics solely from this limited standard suite.



Figure 175 Optical sensor

13.5.2 Kalman Filtering and Kinematic Estimation

Deriving parameters like roll angle purely from a standard IMU is mathematically flawed. Integrating raw gyroscope signals over time introduces a severe "random walk" drift due to inherent sensor bias. For example, over a mere 150-second out-lap, raw integration can yield an estimated roll angle of 78 degrees when the actual lean is only 53 degrees.

To solve this, the Performance Engineering group applies Kalman Filtering techniques. By combining real-time, noisy IMU measurements with a simplified dynamic mathematical model of the motorcycle, the filter successfully washes out the sensor noise and drift. This allows engineers to accurately estimate critical, unmeasurable states—such as roll angle, side slip, and mechanical grip—providing the vital data needed to tune ECU parameters and validate the virtual multi-body models against actual track performance.

14 Seminar: Motorsport technology for cycling tyres (Pirelli)

14.1 Introduction

This seminar offered a detailed examination of the critical characteristics defining high-performance cycling tyres. The lecture explored the specialized manufacturing techniques and structural configurations that enable these tyres to achieve superior performance metrics.



Figure 176

14.2 Anatomy and architecture

A cycling tyre is realized according to a manufacturing process that consists of two main phases.

First, all layers are deposited onto a **steel drum** to form a belt. This semi-finished product undergoes **curing** inside a mould to define its final profile, tread pattern, and geometry. The process concludes with **vulcanization**, a chemical treatment where heat and sulfur create cross-links between polymer chains, converting the rubber into a more durable and elastic structural material.

With regard to the tyre structure, it is composed of the following main layers.

- **Casing:** A structural membrane of **calendered rubberized fabric** that contains air pressure. Its **cross-ply configuration** (40° – 50° angle) dictates dynamics and load capacity, while **chord diameter** and **TPI (Threads per inch)** determine density, tear resistance, and puncture protection.
- **Beads:** Rings of **steel or aramid fiber** (Kevlar) that anchor the tyre to the rim. The material choice affects foldability, balancing a secure fit with the ease of manual mounting.
- **Protective belts:** Under-tread layers of **non-isotropic, square-woven fabric**. These low-modulus, tear-resistant barriers provide puncture protection while significantly impacting rolling resistance and handling.

14.3 Structural mechanics and pressure dynamics

A cycling tyre is a **pure membrane**, possessing no load capacity without inflation and being one order of magnitude lighter than automotive tyres to accommodate limited human power. The **compressed air** is the actual load-bearing element, operating through the **spoke analogy**: pre-stressed casing threads support the rim just as tensioned spokes

suspend a hub.

Due to this thin construction, professional road tyres require pressures of **5–6 bars**, significantly higher than motor vehicles. Pressure requirements exhibit a **quadratic dependency** on tyre width; while road tyres prioritize high-pressure efficiency, off-road disciplines (Gravel/MTB) utilize wider sections up to **60 mm** at lower pressures to navigate rough terrain. Finally, air pressure is essential for **beading**, forcing the beads to lock into the rim flanges and generating a **hoop stress** that exceeds **100 kg**.

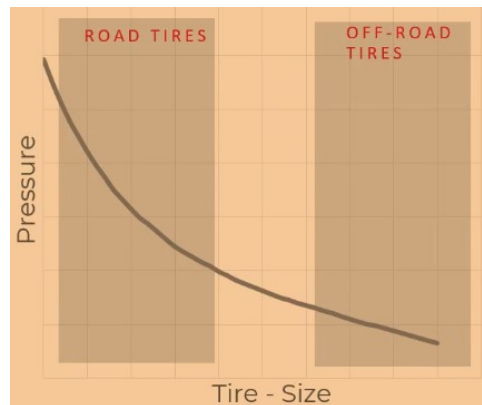


Figure 177: Tyre size - pressure relationship

14.4 Interaction with the ground: Rolling resistance and impedance

The contact patch of a professional cycling tyre is relatively small, measuring approximately **100 mm by 6 mm**. Optimizing how this area interacts with the tarmac is critical for efficiency.

It is a common misconception that higher inflation pressure always leads to lower resistance. In real-world conditions, total energy loss is the sum of two factors:

- **Pure rolling resistance:** The energy dissipated by the deformation of the rubber and casing material.
- **Impedance loss:** The energy lost to vibrations and vertical movement when the tyre is too stiff to absorb road imperfections.

Maximum efficiency is achieved at an **optimal pressure point** where the combination of these two losses is at its lowest.

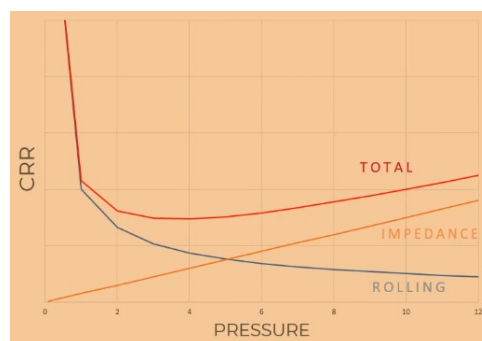


Figure 178: Coefficient of rolling resistance - pressure relationship

14.5 Aerodynamics and system integration

At high racing speeds, particularly in flat time trials, **air drag accounts for more than 50% of the total energy expenditure**. Consequently, the tyre is not treated as an isolated component but as part of a single **airfoil system** alongside the rim. A smooth transition between the tyre and the rim is essential to prevent **airflow separation (stall)**, which causes a significant increase in drag.

Designers utilize **Computational Fluid Dynamics (CFD)** for rapid geometric iterations and **wind tunnels** to validate the absolute drag values of physical prototypes. These aerodynamic optimizations can provide gains of approximately **20 watts**, which is often the margin between a podium finish and the rest of the field.

14.6 Multivariable optimization and trade-offs

Tyre design is a **multivariable problem** where improving one performance target often compromises another. These trade-offs are central to professional race preparation:

- **Width vs. Aerodynamics:** Wider tyres offer better rolling resistance and comfort but increase the frontal area and can disrupt airflow.
- **Weight vs. Protection:** Reducing material lowers weight for climbing but decreases puncture resistance and durability.

Future advancements are heavily focused on **materials research**, specifically new fibers for faster casings and advanced rubber compounds for better grip. While **airless tyres** are emerging for urban micromobility to eliminate punctures, they are currently unsuitable for high-performance racing as they lack the **vibration damping and shock absorption** characteristics of a pressurized pneumatic chamber.

14.7 Impact of Parameters on Race Time

Based on a simulated 40 km time trial (70 kg rider at 390 watts), the impact of different parameters is quantified as follows:

Parameter Varied	Value Change	Power Impact	Time Lost (40km)
Rolling Resistance (Crr)	0.0025 → 0.0035	+10 Watts	+40 seconds
Handling (Wet/Corners)	Loss of confidence	-	+30 seconds
Puncture (Wheel Change)	Quick Stop & Go	-	+50 seconds
Comfort (Efficiency)	Less absorption	-2% Watts	+30 seconds
Aerodynamics (Tire Aero)	9W → 12W @45km/h	+4.1 Watts	+18 seconds
Tyre Weight	250g → 320g	+0.14 kg	+2 seconds

Table 28

- Crr is critical: Rolling resistance is the dominant technical factor, costing 40 seconds.

- Weight is negligible: Despite the 500m climb, tyre weight has almost no impact (+2s) compared to other factors
- Human factors matter: Poor handling and low comfort heavily penalize time (+30s each) by reducing athlete confidence and power efficiency.
- Reliability first: Puncture protection remains a priority, as a wheel change is the single biggest time penalty (+50s)

15 Seminar: Mechanical Design and Mechatronics in Racing Applications (Marelli)

15.1 Introduction

The Marelli Motorsport seminar provides an overview of the **mechanical design and mechatronic integration** required for high-performance racing environments. The lecture focuses on the challenges of protecting sensitive electronics against extreme physical stressors, specifically concentrating on **Electronic Control Units (ECUs)** and **Motor Generator Units (MGUs)**. The design process emphasizes balancing efficiency, weight, and reliability under conditions of high temperature, intense vibration, and environmental contamination.

Key devices analyzed include:

- **ECU:** The complex control system composed of a box, cover, PCBs, and custom spacers.
- **DC/DC Converters:** Devices that convert high battery voltage to lower output tensions (e.g., 48V, GND, 14V).
- **Inverters:** High-power modules that drive the MGU.

15.2 Electronic Control Unit (ECU) and PCB Design

The ECU assembly (build-up) involves a structural integration of a BOX, internal PCBs, board-to-board connectors, and custom spacers. The front cover typically hosts the main signal connectors.

15.2.1 PCB Materials and Layers

Two primary types of PCB substrates are utilized, with specific layering:

- **FR4 PCB:** A multilayer board composed of copper layers and a $50\mu\text{m}$ FR4 core (glass fibers in epoxy resin). It is the standard for insulation, with total thicknesses between 1.6 mm and 4 mm.
- **PCB IMS (Insulated Metal Substrate):** A high-performance stack consisting of a Copper layer, a Dielectric layer, and an **Aluminum Base**. This structure offers excellent heat dissipation and high mechanical strength.

Components are integrated using Surface Mounted Technology (SMT), featuring components like SMT capacitors, inductors, and steel spacers, or Through Hole Technology (THT). **Lead-free tin solder alloys** (melting point $\approx 227^\circ\text{C}$) have been the standard since 2016, requiring specialized fluxes to ensure quality by removing surface oxides.

15.2.2 Connectors and Signal Integrity

Connectors are required to conduct signals between boards (board-to-board), from the ECU to external devices, or for power.

- **Board-to-Board:** High-density solutions like Samtec SMT or Tiger Eye contacts.

- **External/Circular Connectors:** Specialized series like the **Deutsch/Souriau 8STA**. Features include aluminum shells with conductive black zinc plating, liquid silicone rubber seals, and gold-plated copper alloy contacts.
- **Power Connectors:** Designed for high current, such as Amphenol specialized drawings with specific keyway and sealing requirements.

15.2.3 Thermal Management

Thermal management ensures components do not exceed their **Junction Temperature** (T_j). The design goal is to transfer internal heat (from self-heating and external sources) to the external heatsink.

Heat transfer methods:

- **Conduction:** Utilizing Thermal Interface Materials (TIM):
 - **Thermal Grease:** Used for very small thicknesses to compensate for tension (3-5 W/mK).
 - **Thermal Pads:** Silicone-based, thicknesses from 0.5mm to 8mm, with conductivity ranging from **2 W/mK to 20 W/mK**.
- **Convection:** "Active" cooling systems using water/glycol mixtures, essential for power electronics where losses are significant.

Thermal design is based on the first law of Fourier:

$$\dot{Q} = \frac{\lambda}{s} A (T_2 - T_1) \quad (50)$$

- $\dot{Q}$: Thermal power [W]
- λ : Thermal conductivity [W/mK]
- s : Material (wall) thickness [m]
- A : Surface area [m^2]
- $(T_2 - T_1)$: Temperature gradient [K]

Analogous to electrical circuits, the temperature delta is:

$$\Delta T_{AB} = P_T \times \theta_{AB} \quad (51)$$

- ΔT_{AB} : Temperature difference [K]
- P_T : Thermal power [W]
- θ_{AB} : Thermal resistance [K/W]

The total thermal path for a component to ambient is:

$$T_j - T_a = (R_{\theta j-p} + R_{\theta TIM} + R_{\theta hs-a}) \times P_T \quad (52)$$

- $R_{\theta j-p}$: Junction-to-Package resistance [K/W]
- $R_{\theta TIM}$: TIM resistance [K/W]
- $R_{\theta hs-a}$: Heatsink-to-Ambient resistance [K/W]

15.2.4 Vibration and Environmental Management

Vibrations (random or shock) cause fatigue and breakage. Analysis involves simulation to predict effects on PCB components and layout guidelines to minimize resonance frequencies.

Vibration Classes:

- **VA/VB:** F1 environment (Engine/Car).
- **VC/VD:** WEC/Cockpit (Halo, Dashboards).
- **VE/VT:** Motorcycles/Ground applications.

Environmental Sealing and IP Protection: To prevent electrical shorts (IP classification tests like IP62), Marelli uses:

- **Sealing/Potting:** Motorblack gaskets, silicone, and HUMISEAL coatings.
- **IP Air Tool Test:** Quality check using air flow (40 mbar for 300 sec, max delta pressure 5 mbar).
- **Pressure Balance Valves:** Automotive vent valves (snap-fit, screwed, or adhesive) to equalize pressure during temperature changes.

15.3 MGU and Future Technological Trends

The MGU system converts DC battery current to AC. This technology has seen a radical evolution in F1 Hybrid Architectures:

- **2009-2013:** 60kW ERS-K.
- **2014-2025:** 120kW ERS-K + 125krpm ERS-H.
- **2026:** 350kW ERS-K (minimum weight 16kg).

Power Module Evolution:

- **2008 (IGBT):** 60kW, Power Density 136 kW/l.
- **2020 (2nd Gen SiC):** 200kW, Power Density 1000 kW/l.
- **2026 (3rd Gen SiC):** Transitioning to full Silicon Carbide MOSFET systems.

15.3.1 Design Optimization and Future Challenges

Designers use a proprietary "Optimizer" master software to run Python scripts for electromagnetic, mechanical, and thermal analyses, searching for **Pareto optimal solutions** that minimize weight and volume while maximizing efficiency.

Future challenges for increasing power/volume ratios include:

- **Passive Vapor Chambers** for localized heat spread.
- **Active cooling channels** integrated directly into the PCB using additive metal manufacturing.